

Linear System Fall 2026

Homework Three

Due: Friday, Oct 16

Exercise 1 Obtain the transfer function of the following system

$$\begin{aligned}\dot{x}(t) &= -x(t) + 2x(t-h) + u(t), \\ y(t) &= x(t).\end{aligned}$$

Exercise 2 Obtain a state space realization (controllable canonical form) of the transfer function

$$\hat{G}(s) = \frac{s^2 + 3s + 2}{s^2 + 5s + 6}.$$

Is your realization minimal?

Exercise 3 Obtain a state space representation (controllable canonical form) of the following transfer function:

$$\hat{G}(s) = \begin{pmatrix} \frac{s^2 + 1}{s^2 - 1} & \frac{2}{s - 1} \end{pmatrix}.$$

Exercise 4 Obtain a state space realization (controllable canonical form) of the transfer function

$$\hat{G}(s) = \begin{pmatrix} \frac{1}{s - 1} & \frac{1}{s - 1} \\ \frac{1}{s + 1} & -\frac{1}{s + 1} \end{pmatrix}.$$

Exercise 5

(a) Obtain a state space realization (controllable canonical form) of the following single-input single-output system:

$$\ddot{y} - 3\dot{y} - 4y = \ddot{u} - 2\dot{u} - 8u.$$

(b) Is your realization minimal?

Exercise 6 Obtain a state space (controllable canonical form) realization of the following input-output system:

$$\begin{aligned}\dot{y}_1 + y_2 &= \dot{u}_2 + u_1, \\ \dot{y}_2 + y_1 &= \dot{u}_1 + u_2.\end{aligned}$$

Exercise 7 Determine whether or not the following system of linear equations has a solution. If a solution exists, determine whether or not it is unique, and if not unique, obtain an expression for all solutions.

$$\begin{aligned}x_1 - x_2 + 2x_3 + x_4 &= 5, \\x_1 - x_2 + x_3 &= 3, \\-2x_1 + 2x_2 + 2x_4 &= -2, \\2x_1 - 2x_2 - x_3 - 3x_4 &= 0.\end{aligned}$$

Exercise 8 We consider here systems of linear equations of the form

$$Ax = b$$

with

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

For each of the following cases, we present the reduced row echelon form of the augmented matrix $[A \ b]$. Determine whether or not the corresponding system of linear equations has a solution. If a solution exists, determine whether or not it is unique; if not unique, obtain an expression for all solutions and give two solutions.

$$\begin{array}{ll} \text{(a)} \quad \begin{pmatrix} 1 & 5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \text{(b)} \quad \begin{pmatrix} 1 & -2 & 0 & 6 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ \\ \text{(c)} \quad \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} & \text{(d)} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \end{array}$$

Exercise 9 Consider the three vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}.$$

(a) Express the vector

$$\mathbf{v} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$$

as a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

(b) Can every vector of the form

$$\mathbf{v} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

be expressed as a **linear combination** of $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$?

(c) Are the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$ linearly independent?

Exercise 10

(a) Find a basis for the **null space** of the matrix,

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{pmatrix}.$$

(b) What is the **nullity** of A ?

Exercise 11 (a) Find a basis for the **range** of the matrix,

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{pmatrix}.$$

(b) What is the **rank** of A ?

Exercise 12

Compute the **eigenvalues** and **eigenvectors** of the matrix,

$$A = \begin{pmatrix} 2 & -3 & 0 \\ 2 & -3 & 0 \\ 3 & -5 & 1 \end{pmatrix}.$$

Exercise 13

Compute the **eigenvalues** and **eigenvectors** of the following matrix.

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$